Final Project Report

Name: Kavita Annaji Mangalkar

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Final Project Submission

I. Overview

The Weather Forecast Web Application is a web-based system developed using Java Servlet, HTML, and CSS that provides real-time weather information for any city. The application integrates with the OpenWeather API to fetch live weather data such as temperature, humidity, wind speed, and weather conditions.

The main objective of this project is to demonstrate how web technologies and API integration can be used to build dynamic and interactive applications. The system allows users to enter a city name and retrieve accurate weather details instantly. It follows a client-server architecture where the front end interacts with the backend servlet, which processes requests and fetches data from the API.

II. Objectives

- To develop a web-based weather forecasting system
- To implement API integration using OpenWeather API
- To understand Java Servlet for backend processing
- To display real-time weather information dynamically
- To create a user-friendly interface

III. System Requirements

- Hardware Requirements:
  - Computer/Laptop
  - Minimum 4GB RAM
- Software Requirements:
  - Java (JDK 8 or above)

- Apache Tomcat Server
- Eclipse IDE
- Web Browser (Chrome/Edge)
- Internet connection (for API access)

IV. Technologies Used

- Frontend: HTML, CSS
- Backend: Java Servlet
- API: OpenWeather API
- Server: Apache Tomcat
- IDE: Eclipse

V. System Architecture

The system follows a 3-tier architecture:

1. Presentation Layer (Frontend):
    - User enters city name through HTML form
2. Application Layer (Servlet):
    - Processes user request
    - Sends request to OpenWeather API
    - Parses JSON response
3. Data Layer (API):
    - OpenWeather API provides real-time weather data

VI. Methodology

1. Requirement Analysis
    - Identify user needs (search weather by city)
2. System Design
    - Design UI using HTML & CSS
    - Plan servlet workflow
3. API Integration
    - Connect OpenWeather API using HTTP request
    - Fetch JSON data
4. Implementation

    - Develop servlet for processing request
    - Parse and display data
5. Testing
    - Test for different city inputs
    - Handle invalid inputs
6. Deployment
    - Deploy on Apache Tomcat server

VII. Features of the System

- Search weather by city name
- Real-time weather updates
- Displays temperature, humidity, and conditions
- User-friendly interface
- Fast response using API

VIII. Working of the System

1. User enters a city name in the input field
2. Request is sent to the Java Servlet
3. Servlet calls the OpenWeather API
4. API returns weather data in JSON format
5. Servlet processes and extracts required data
6. Data is displayed on the web page

IX. Advantages

- Provides real-time weather information
- Simple and easy to use
- Demonstrates API integration
- Lightweight web application

X. Limitations

- Requires internet connection
- Depends on API availability
- Limited to current weather data only

XI. Future Scope

- Add 7-day weather forecast
- Use GPS for automatic location detection
- Improve UI using Bootstrap
- Add weather icons and animations
- Develop mobile application version

XII. Conclusion

The Weather Forecast Web Application successfully demonstrates the use of Java Servlet and API integration to build a dynamic web application. It provides accurate and real-time weather data with a simple interface. This project helps in understanding web development concepts, server-side programming, and external API usage.

XIII. References

- OpenWeather API Documentation
- Java Servlet Documentation
- Apache Tomcat Official Website